

Samanvay Anand

PhD Scholar, Department of Aerospace Engineering,
Indian Institute of Science, Bengaluru-560012, India

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RESEARCH INTERESTS

- Wave Propagation
- Mechanical Metamaterials
- Magneto Active Polymers
- Solid Mechanics

EDUCATION

Indian Institute of Science (IISc), Bengaluru 2020 - Present
Doctor of Philosophy (PhD), Aerospace Engineering. CGPA: 8.3/10.00
Supervisors: Prof. Rajesh Chaunsali and Prof. Vivekanand Dabade

Indian Institute of Technology (IIT), Mandi 2018 - 2020
Master of Technology (MTech), Mechanical Engineering – Energy Systems. CGPA: 8.5/10.00
Supervisor: Prof. Himanshu Pathak

Visvesvaraya Technological University (VTU), Belagavi 2012 - 2016
Bachelor of Engineering (BE), Mechanical Engineering. Percentage: 74.77/100
Supervisor: Prof. Gururaja Sharma T

AWARDS AND SCHOLARSHIPS

Ministry of Education (MoE) Fellowship 2020 - Present
Awarded for pursuing Doctoral research at IISc, Bengaluru by the Ministry of Education, Government of India.

Ministry of Human Resource Development (MHRD) Fellowship 2018 - 2020
Awarded for pursuing a Master's degree at IIT, Mandi after qualifying for the Graduate Aptitude Test in Engineering (GATE) 2018, by the Ministry of Human Resource Development, Government of India.

RESEARCH PUBLICATIONS

Journal Publications

1. Ray, A. *, **Anand, S. ***, Dabade, V. and Chaunsali, R., 2025. Remote nucleation and stationary domain walls via transition waves in tristable magnetoelastic lattices. *Physical Review Materials*, 9(1), p.014405.
2. Suman, S., Dwivedi, K., **Anand, S.** and Pathak, H., 2022. XFEM-ANN approach to predict the fatigue performance of a composite patch repaired aluminium panel. *Composites Part C: Open Access*, 9, p.100326.

**Equally contributed first author*

Conference Proceedings

1. **Anand, S.** and Pathak, H., 2020, March. Finite element modelling of in situ composite patch repair of cracked aluminium aircraft structures. In *Indian Structural Steel Conference* (pp. 243-259). Singapore: Springer Nature Singapore.

OTHER SCHOLARLY ACTIVITY

Conferences

Oral presentations

1. "Transition waves and collision dynamics in tunable multistable mechanical metamaterials", ASME-IMECE, 16 - 20 November 2025, Memphis, Tennessee, USA.
2. "Transition waves and their collision dynamics in multistable magnetoelastic lattices", ICMAMS, 11 - 13 December 2024, Bengaluru, India.
3. "Nonlinear waves in multistable magnetoelastic lattices", SAE AeroCON, 6 - 7 June 2024, Bengaluru, India.

Symposiums

Oral presentations

1. “A tunable design approach for transition wave propagation”, Aerospace Research Students’ Symposium (AERES), 28 - 31 January 2025, IISc, Bengaluru, India.
2. “Nonlinear waves in multistable magnetoelastic lattices”, AERES, 3 - 6 January 2024, IISc, Bengaluru, India.

Poster presentation

1. “Collision dynamics of transition waves in a tristable magnetoelastic lattices propagation”, Mechanical Divisional Symposium, 10 - 11 May 2024, IISc, Bengaluru, India.

TEACHING

- **Tutor**, AE 296 - Experimental Methods in Aerospace Engineering, August - December 2024 and 2025, IISc, Bengaluru, India.
- **Teaching Assistant**, AE 204A - Mechanics of Flight Vehicle Structures, August - December 2024, IISc, Bengaluru, India.
- **Teaching Assistant**, Product Realization Technology Lab, February - July 2020, IIT Mandi, India.
- **Teaching Assistant**, Mechanics of Rigid Bodies, August - November 2019, IIT Mandi, India.
- **Teaching Assistant**, Design Practicum, February - June 2019, IIT Mandi, India.
- **Teaching Assistant**, Central Library, August - November 2018, IIT Mandi, India.

SERVICE AND LEADERSHIP ACTIVITIES

- **Organizer**, CISM-IISc Workshop, 21 - 25 July 2025, Dept. of Aerospace Engineering, IISc, Bengaluru, India.
- **Organizer**, AERES, 3 - 6 January 2024, Dept. of Aerospace Engineering, IISc, Bengaluru, India.
- **Organizer**, AERES, 17 - 20 January 2023, Dept. of Aerospace Engineering, IISc, Bengaluru, India.
- **Student Member**, Mountain Biking Club, 2018 - 2020, IIT, Mandi, India.
- **Student Member**, Indian Society of Heating Refrigeration and Air Conditioning Engineers (ISHRAE), 2015 - 2016, India.

INDUSTRY EXPERIENCE

Industrial Training

- Studied the working and operation of “Tunnel Boring Machine” at **Tapovan Vishnugad Hydro Power Plant (TVHPP), NTPC Ltd.**, June - July 2019, Joshimath, Uttarakhand, India.

TECHNICAL SKILLS

- **CAD and FEM Softwares**: CATIA, ABAQUS, MSC Patran-Nastran, COMSOL.
- **Programming and Scripting Languages**: C++, MATLAB, Python, LabVIEW.

OTHER ACTIVITIES

Workshops

- **CISM-IISc Workshop**, 21 - 25 July 2025, Dept. of Aerospace Engineering, IISc, Bengaluru, India.
- **Science Communication and Grant Writing Workshop**, 6 February 2025, Office of Research Grants, IISc, Bengaluru, India.
- **Scientific Writing Workshop**, 31 March 2021, ISRO, India.
- **Material Characterization Techniques**, 16 - 17 September 2020, IIT Patna, India.
- **Computational Methods in Mechanical Engineering**, 3 - 7 March 2020, NIT Hamirpur, India.
- **Water Filtration Technique Based on Solar Energy**, 11 - 13 February 2019, IIT Mandi, India.